Del principio de acción al problema de Sturm–Liouville

Desarrollo simbólico

1. Acción y ecuación de Euler–Lagrange

1. Acción.

$$L \equiv T - V, \qquad S[q] = \int_{t_1}^{t_2} \mathrm{d}t\; L(q, \dot{q}) \tag{1.1}$$

2. Variación.

$$\mathrm{d}f = \frac{\partial f}{\partial x_i}\,\mathrm{d}x_i = 0 \quad \longleftrightarrow \quad \delta S = \varepsilon\,\frac{\mathrm{d}}{\mathrm{d}\varepsilon} S[q+\varepsilon\eta]\Big|_{\varepsilon=0} = 0, \qquad \delta q(t_1) = \delta q(t_2) = 0 \tag{1.2}$$

3. Euler–Lagrange.

$$\delta S = \int_{t_1}^{t_2} \mathrm{d}t \left[\frac{\partial L}{\partial q}\,\delta q + \frac{\partial L}{\partial \dot{q}}\,\delta\dot{q}\right] \tag{1.3}$$

$$= \underbrace{\left[\frac{\partial L}{\partial \dot{q}}\,\delta q\right]_{t_1}^{t_2}}_{=0} + \int_{t_1}^{t_2} \mathrm{d}t \left[\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}}\right]\delta q = 0 \quad \forall\,\delta q \tag{1.4}$$

$$\Longrightarrow \quad \boxed{\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = 0} \tag{1.5}$$

4. Partícula en un potencial.

$$L = \tfrac{1}{2}\,m\,\dot{x}^2 - V(x) \tag{1.6}$$

$$\delta S = \int_{t_1}^{t_2} \mathrm{d}t \left[\tfrac{1}{2}\,m\,2\dot{x}\,\delta\dot{x} - V'(x)\,\delta x\right] = \int_{t_1}^{t_2} \mathrm{d}t \left[m\,\tfrac{\mathrm{d}}{\mathrm{d}t}(\dot{x}\,\delta x) - m\ddot{x}\,\delta x - V'(x)\,\delta x\right] \tag{1.7}$$

$$= \underbrace{\left[m\,\dot{x}\,\delta x\right]_{t_1}^{t_2}}_{=0} + \int_{t_1}^{t_2} \mathrm{d}t\,[-m\ddot{x} - V'(x)]\delta x \tag{1.8}$$

$$\Longrightarrow \quad -m\ddot{x} - V'(x) = 0 \quad \Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}t}\Big(\tfrac{1}{2}m\dot{x}^2 + V(x)\Big) = \dot{x}\,(m\ddot{x} + V'(x)) = 0 \tag{1.9}$$

2. Lagrangiano cuadrático

1. Linealización. $V'(x_0) = 0,\ V''(x_0) > 0,\ \xi = x - x_0$:

$$V(x) = V(x_0) + \tfrac{1}{2}V''(x_0)\,\xi^2 + O(\xi^3) \implies L \simeq \tfrac{1}{2}m\dot{\xi}^2 - \tfrac{1}{2}V''(x_0)\,\xi^2 \implies m\ddot{\xi} = -V''(x_0)\,\xi \tag{2.1}$$

2. Acción de campo.

$$S[\psi] = \int_{t_1}^{t_2}\mathrm{d}t \int_\Omega \mathrm{d}^3x \left[\tfrac{1}{2}\,k(\mathbf{x},t)\,\dot{\psi}^2 - \tfrac{1}{2}\,u(\mathbf{x},t)\,|\nabla\psi|^2 - \tfrac{1}{2}\,M(\mathbf{x},t)\,\psi^2\right], \qquad k>0,\ u>0,\ M \text{ acotada} \tag{2.2}$$

3. Variación. $\delta\psi = 0$ en t_1, t_2 y $\partial\Omega$:

$$\delta S = \int \mathrm{d}t\,\mathrm{d}^3x\ [k\,\dot{\psi}\,\delta\dot{\psi} - u\,\nabla\psi\cdot\nabla\delta\psi - M\,\psi\,\delta\psi] \tag{2.3}$$

$$= \int \mathrm{d}t\,\mathrm{d}^3x \left[\partial_t(k\dot{\psi}\,\delta\psi) - \partial_t(k\dot{\psi})\delta\psi - \nabla\cdot(u\nabla\psi\,\delta\psi) + \nabla\cdot(u\nabla\psi)\delta\psi - M\psi\,\delta\psi\right] \tag{2.4}$$

$$= \int \mathrm{d}t\,\mathrm{d}^3x \left[-\partial_t(k\,\dot{\psi}) + \nabla\cdot(u\,\nabla\psi) - M\,\psi\right]\delta\psi \tag{2.5}$$

$$\Longrightarrow \quad \boxed{-\partial_t(k\,\partial_t\psi) + \nabla\cdot(u\,\nabla\psi) - M\,\psi = 0} \qquad \Big(\partial_t\frac{\partial\mathcal{L}}{\partial\dot{\psi}} + \nabla\cdot\frac{\partial\mathcal{L}}{\partial(\nabla\psi)} - \frac{\partial\mathcal{L}}{\partial\psi} = 0\Big) \tag{2.6}$$

3. Problema de Sturm–Liouville

1. Una dimensión, sistema estático. $k = k(x),\ u = u(x),\ M = M(x)$:

$$-k(x)\,\frac{\partial^2\psi}{\partial t^2} + \frac{\partial}{\partial x}\Big[u(x)\,\frac{\partial\psi}{\partial x}\Big] - M(x)\,\psi = 0 \tag{3.1}$$

2. Separación. $\psi(x,t) = T(t)\,X(x)$:

$$-X\,\frac{\mathrm{d}^2T}{\mathrm{d}t^2} + \frac{1}{k}\Big[\frac{\mathrm{d}}{\mathrm{d}x}\Big(u\frac{\mathrm{d}X}{\mathrm{d}x}\Big) - M\,X\Big]T = 0 \tag{3.2}$$

$$-\frac{1}{T}\frac{\mathrm{d}^2T}{\mathrm{d}t^2} + \frac{1}{k\,X}\Big[\frac{\mathrm{d}}{\mathrm{d}x}\Big(u\frac{\mathrm{d}X}{\mathrm{d}x}\Big) - M\,X\Big] = 0 \quad\Longrightarrow\quad -\frac{1}{T}\frac{\mathrm{d}^2T}{\mathrm{d}t^2} = \lambda = \frac{1}{k\,X}\Big[-\frac{\mathrm{d}}{\mathrm{d}x}\Big(u\frac{\mathrm{d}X}{\mathrm{d}x}\Big) + M\,X\Big] \tag{3.3}$$

$$\boxed{\frac{\mathrm{d}^2T}{\mathrm{d}t^2} = -\lambda\,T, \qquad \frac{1}{k(x)}\Big[\frac{\mathrm{d}}{\mathrm{d}x}\Big(-u(x)\frac{\mathrm{d}X}{\mathrm{d}x}\Big) + M(x)\,X\Big] = \lambda\,X} \tag{3.4}$$

3. Operador de Sturm–Liouville.

$$\mathcal{L}_{\mathrm{SL}} = \frac{1}{k}\Big[-\frac{\mathrm{d}}{\mathrm{d}x}\Big(u\,\frac{\mathrm{d}}{\mathrm{d}x}\Big) + M\Big], \qquad -(pX')' + qX = \lambda\,w\,X, \quad p = u,\ q = M,\ w = k \tag{3.5}$$

$$\lambda > 0 \implies T(t) = A\cos(\sqrt{\lambda}\,t) + B\sin(\sqrt{\lambda}\,t), \qquad \omega = \sqrt{\lambda} \tag{3.6}$$

Apéndice: Legendre en $x \to \infty$

$$(1-x^2)\,y'' - 2x\,y' + n(n+1)\,y = 0, \quad n \in \mathbb{N}; \qquad y = \sum_{k\geq 0} c_k\,x^{-(r+k)} \tag{A.1}$$

$$r^2 - r - n(n+1) = 0 \implies r = n+1,\ \ r = -n; \qquad c_k = \frac{(r+k-2)(r+k-1)}{(r+k-n-1)(r+k+n)}\,c_{k-2}, \quad c_1 = 0 \tag{A.2}$$

$$r = n+1: \quad a_k = \frac{(k+n-1)(k+n)}{k(k+2n+1)}\,a_{k-2} \neq 0 \ \ \forall k \implies Q_n(x) = x^{-(n+1)}\sum_{k\ \mathrm{par}} a_k x^{-k} \tag{A.3}$$

$$r = -n: \quad b_k = \frac{(k-n-2)(k-n-1)}{k(k-2n-1)}\,b_{k-2}, \quad k - 2n - 1 \neq 0 \ \ (k\ \mathrm{par}) \tag{A.4}$$

$$n\ \mathrm{par}:\ b_{n+2} = 0, \qquad n\ \mathrm{impar}:\ b_{n+1} = 0 \implies P_n(x) = x^n \sum_{\substack{k\ \mathrm{par}\\ k\leq n}} b_k x^{-k} \tag{A.5}$$

$$Q_n:\ |x| > 1, \qquad P_n:\ x \in \mathbb{R}\ (\rho = \infty) \tag{A.6}$$

$$((1-x^2)y')' + n(n+1)\,y = 0 \ \ (p = 1-x^2,\ q = 0,\ w = 1,\ \lambda = n(n+1)) \tag{A.7}$$